Toxic Text Classification Using Deep Learning Techniques

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***Abstract*—Recent growth in social media platforms has made automated toxic text classification essential. In this paper, we train and evaluate three models, Logistic Regression, LSTM, and DistilBERT on a binary text classification task using an augmented version of the Jigsaw Toxic Comment Classification dataset. The augmented dataset contains 178,614 comments with a 20.3% toxic rate. Among the models evaluated, DistilBERT achieved the best results with a ROC-AUC of 0.9727 and an F1-score of 0.71 on the toxic class at a 0.5 threshold. This work highlights the impact of class imbalance affect the model prediction result and how threshold selection can help balance precision and recall in imbalanced datasets.**

Keywords—Deep Learning, NLP, Text Classification, LSTM, Transformers Models

I. Introduction

The rapid growth of social media has contributed to the widespread proliferation of toxic content on the internet, such as hate speech, harassment, and offensive comments. Online platforms such as Twitter and Facebook process millions of posts and comments daily, making manual review impractical for such a scale. This has created the need for automated intelligent systems that are capable of detecting toxic text.

Previous approaches to text classification, such as Logistic Regression and other traditional machine learning algorithms, relied on the use of handcrafted features along with basic text representations such as bag-of-words or TF-IDF. Although such approaches are computationally efficient, they generally do not focus on the contextual meaning of the texts, especially when the texts involve complex situations such as sarcasm, implicit hate speech, or social media language. Deep learning Techniques such as LSTM [1] was proposed to efficiently deal with sequential data such as texts, as they can easily understand the relationships between the words present in a sentence. And recently Transformers-based [2] models such as BERT [3] can understand the contextual relationships between the words present in a sentence or a piece of text by using the attention mechanism thus allowing the models to perform better on various text classification tasks such as the detection of toxic comments.

Despite these advancements, one of the key challenges in toxic text classification remains in class imbalance, where non-toxic samples significantly outnumber toxic ones. This imbalance can bias models toward the majority class, resulting in poor detection of toxic content. Addressing this issue is essential for building reliable and fair classification systems.

In this paper, we will focus on a binary text classification task that will involve the training and evaluation of three different models - Logistic Regression, LSTM, and DistilBERT [4] on an augmented version of the Jigsaw Toxic Comment Classification Dataset [5]. Our main objectives are:

- Preprocess and analyze text data for toxicity detection.
- Train and implement various machine learning and deep learning models.
- Test and compare the performance of the models using relevant metrics.

The findings of the study will help to gain insights into the efficiency of the modeling techniques for the classification of the toxic comments and the advantages of the transformer model for the complex language understanding task.

II. Related Work

Toxic content detection has evolved significantly in the past few years from traditional machine learning models to complex deep learning approaches and towards advancements to LLMs based detectors. Table I summarizes some of the recent contributions in the field.

TABLE I. SUMMARY OF RELATED WORK

Paper Name	Methods	Technology	Dataset	Results
Lived Experience Matters: Automatic Detection of Stigma on Social Media Toward People Who Use Substances [6]	Jury Learning Framework Deep & Cross Network (DCN) Differential Language Analysis (DLA) Majority Voting with Thresholds	Logistic Regression Extra Trees Classifier Unigrams Tokenization LIWC (Linguistic Inquiry and Word Count) BERTweet	5,000 public Reddit posts	The results show that those with lived experience perceive more stigma, those with lived experience do not always agree on what is stigmatizing, and there is overlap between those with lived experience and those without in what words are more likely to be stigmatized
Fine-Tuning Llama 2 for Detecting Online Sexual Predatory Chats [7]	LoRA (Low-Rank Adaptation) Fine-tuning with Cross-Entropy Loss	Llama 2 7B model AdamW Optimizer Rotary Positional Embeddings (RoPE)	PAN 2012 (English), Roman Urdu (10k), and Urdu Dataset	Achieved 1.00 accuracy and 0.98 f1-score on the PAN12 dataset
A Text-to-Text Model for Multilingual Offensive Language Identification [8]	Text -to-Text Learning	T5-Large/mT5 large models	SOLID 9m tweets and CCTK 1.8m posts	The results demonstrate that the multilingual model outperforms other

				transformer-based models fine-tuned for offensive language detection, such as fBERT and HateBERT
Hate Speech and Offensive Content Detection in Indo-Aryan Languages: A Battle of LSTM and Transformers [9]	Fine-tuning large multilingual models and LSTM	LSTM, mBERT, DistilBERT, XLM-R	HASOC 2023 datasets for Bengali, Assamese, Bodo, Sinhala, and Gujarati	mBERT strongest for Bodo (0.83 F1); XLM-R for Sinhala (0.83); LSTM for Gujarati (0.76)
Assessing the Level of Toxicity Against Distinct Groups in Bangla [10]	fine-tuning transformer models	BanglaBERT Bangla-bert-base mBERT, DistilBERT	3,100 labeled Bangla comments from TikTok, Facebook, and Instagram	Bangla-BERT surpasses alternative models, achieving an F1-score of 0.8903
Detection of Homophobia & Transphobia in Dravidian Languages [11]	implementation of deep learning (CNN, LSTM) and transformer models	CNN, LSTM with GloVe, mBERT, and IndicBERT	DravidianLangTech Malayalam and Tamil datasets.	dicBERT outperforms the other implemented models with obtained weighted average F1-score of 0.86 and 0.77 for Malayalam and Tamil, respectively.
A Review of Hate Speech Detection: Challenges and Innovations [12]	review of rules-based, ML, and deep learning approaches	Lexicons, LR, SVM, Random Forests, CNN, LSTM, and BERT/Transformer variants	Summaries of other papers. (no training dataset)	Reviewed multiple techniques and recommend adopting a multidisciplinary approach
Hate Speech Detection with Generalizable Target-aware Fairness [13]	A new method named “GetFair” that uses Hypernetwork parameter generation, adversarial training, and imitation learning	BERT encoder	Jigsaw Dataset MHS Dataset	Achieved higher metrics scores in effectiveness and fairness than other methods.
Moderating New Waves of Online Hate (HATEGUARD) [14]	leverages the chain-of-thought (CoT) prompting technique	GPT-4 KeyBERT NLTK Wordnet	Self-collected 4000 tweets from covid, capitol insurrection and Ukraine invasion	Higher metrics than existing tools with an accuracy of 0.99, recall 0.99, precision 0.96
Identifying Hate Speech in Telugu Code-Mixed: A BERT Multilingual [15]	Fine-tuning a multilingual model (BERT)	BERT	4,000 Telugu sentences	Macro F1-score of 0.6151

III. DATASET

A. Jigsaw Toxic Comment Classification Dataset

The Jigsaw Toxic Comment Classification Dataset is the primary dataset in this paper. It is the main dataset in the Kaggle challenge made by google and jigsaw. The training dataset consists of 159571 Wikipedia talk page comments with a toxic rate of 10.2%. The original dataset is labeled in six binary toxicity categories: toxic, severe toxic, obscene, threat, insult, and identity hate.

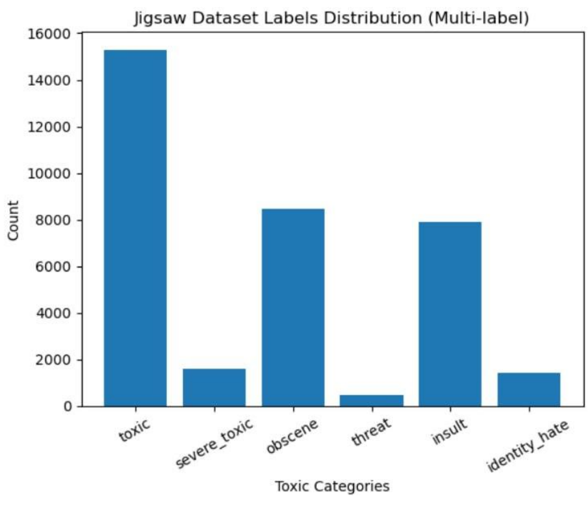

Fig. 1. Original Jigsaw Dataset Distribution.

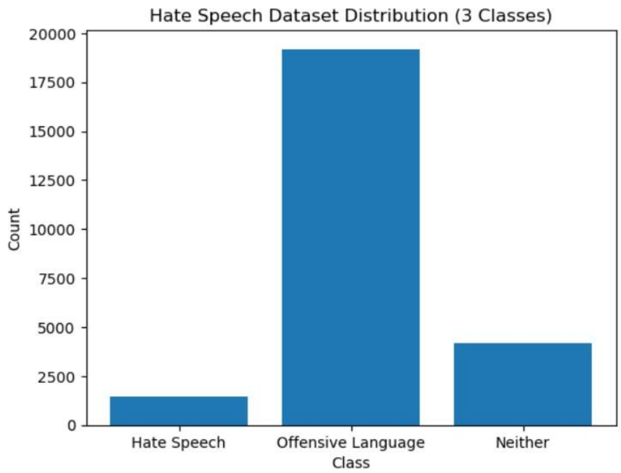

Fig. 2. Original Hate Speech Dataset Distribution.

B. Data Augmentation

Due to class imbalance, we augmented the Jigsaw training set with additional toxic comments from the “Hate Speech and Offensive Language [16]” dataset. Which contains twitter data. The augmented dataset contains 178,614 comments with a toxic rate of 20.3%. The six toxicity labels were merged into a single label to simplify binary classification.

TABLE II. DATASET SPLITS

Split	Total	Toxic
Training	178,614	36,203
Testing	63,978	6,243

IV. METHODOLOGY

A. Preprocessing

All comments went through a cleaning process that involved the removal of mentions and tags, URLs, special characters and punctuation, and special html entities. The comments were converted to lowercase, and any rows with missing values or duplicates were removed.

B. Models

Three different models were trained and evaluated in this paper:

1) Logistic Regression: Logistic Regression is a classical machine learning model used as a baseline. In this work text will be represented using TF-IDF. The model is fast and computationally efficient. However, it has limited capacity to understand the context.

2) LSTM: LSTM is a deep learning model designed for sequential data. They use word embeddings to represent text and they can capture sequential dependencies. However, they find it difficult to understand long-range context.

3) DistilBERT: DistilBERT is a transformer-based model that is trained using knowledge distillation from BERT. It captures deep contextual relationships between words and deliver good performance in text classification tasks. However, it is more computationally demanding than other traditional models.

C. Training Setup

All models were trained locally on two different devices. Logistic Regression was trained using the Scikit-learn library on a core i7-8565U laptop, while LSTM was trained on the TensorFlow library on the same device. DistilBERT was trained on an RTX 5060TI PC using Transformers' library Trainer class.

Different threshold values such as 0.5, 0.7, and 0.75 were applied to the models to deliver better performance. Adjusting the threshold helped reduce false positives and improve precision and F1-score.

TABLE III. SETTINGS/PARAMETERS PER MODEL

Setting/ Parameter	Logistic Regression	LSTM	Distil BERT
Epochs	-	3	1
Batch Size	1000 max iters	32	128
Learning Rate	-	Adam default	2e-5
Training Time	seconds	10m/epoch	12m/epoch
Text Format	TF-IDF Vectors	Embeddings	Tokenized and Padded (max length 256)
Library	Scikit-learn	TensorFlow	Transformers

D. Evaluation Metrics

Model performance is evaluated using four main metrics: precision, which is how accurate the predictions for toxic comments were; recall, which is how much of the toxic comments did the model catch; F1-score, which is a balance between precision and recall; and accuracy, which is the overall correctness of the model.

V. RESULTS

The experimental results show that there are differences in model performance. The results show that both Logistic Regression and LSTM had close results while DistilBERT performed better compared to both models. It is also observed that the model performance is sensitive to the classification threshold. Increasing the threshold improves the precision but lowers the recall. And decreasing the threshold increases the recall at the cost of precision. As shown in Table IV is each model metrics on toxic label using different thresholds.

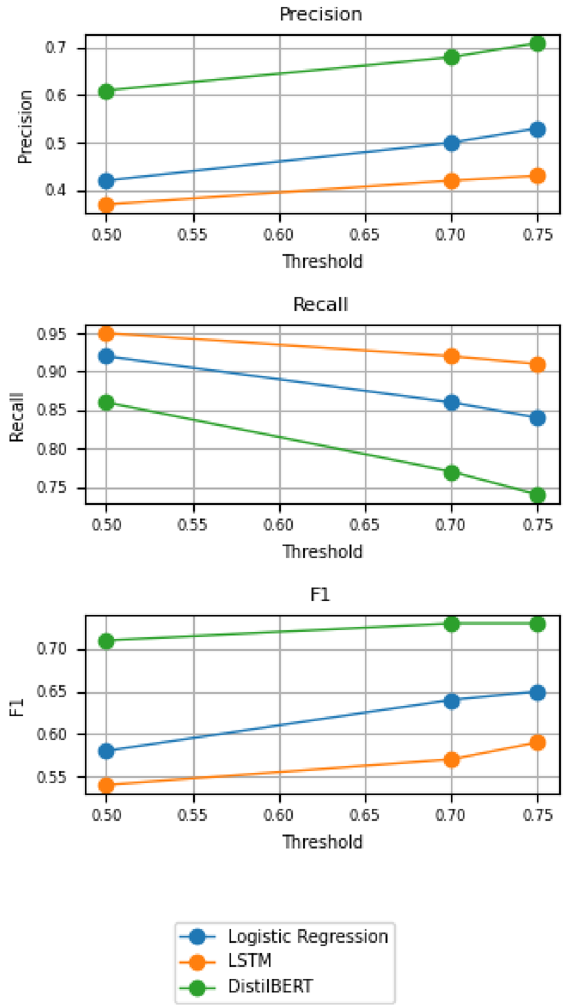

Fig. 3. Models Performance Metrics.

TABLE IV. MODELS PERFORMANCE COMPARISON ON TOXIC

Model	Threshold	Accuracy	Precision	Recall	F1
Logistic Regression	0.50	0.8679	0.42	0.92	0.58
Logistic Regression	0.70	0.9032	0.50	0.86	0.64
Logistic Regression	0.75	0.9116	0.53	0.84	0.65
LSTM	0.50	0.8418	0.37	0.95	0.54
LSTM	0.70	0.8722	0.42	0.92	0.57
LSTM	0.75	0.8809	0.43	0.91	0.59
DistilBERT	0.50	0.9317	0.61	0.86	0.71
DistilBERT	0.70	0.9430	0.68	0.77	0.73
DistilBERT	0.75	0.9449	0.71	0.74	0.73

VI. DISCUSSION

A. Logistic Regression VS LSTM

The performance of Logistic Regression and LSTM models was almost identical. Although the former is simple linear model and the latter is more complex deep learning approach, it seems that LSTM does not really provide a significant performance boost. Which can be explained in a number of ways. First, it is possible that the LSTM model does not really capture the long-range relationships in the comments, which can be quite long. Second, it is also possible that the imbalanced nature of the problem dataset has a similar effect on both models.

It is also worth noting that even a simple Logistic Regression model, when applied properly, can perform quite well in text classification tasks. The problem, however, which both models have, is the low precision on the toxic class, suggesting many false positives.

B. DistilBERT Performance

DistilBERT performed better than the other models across all evaluation parameters. It delivered a strong balance between precision and recall, hence yielding the best F1 score compared to other models. Moreover, the model recorded a high ROC-AUC score of 0.9727, indicating its excellent ability to differentiate between toxic and non-toxic comments.

C. Impact of Class Imbalance

All models were impacted by the data imbalance which resulted in high accuracy scores which does not reflect the real model performance. It also resulted in lower precision for the toxic class due to overprediction. However, DistilBERT outperformed other models with a better balance between precision, recall, and accuracy.

VII. CONCLUSION

In this work, we trained and presented a comparison between three different machine learning and deep learning models- Logistic Regression, LSTM, and DistilBERT on a binary toxic text classification task on an augmented version of Jigsaw dataset that contains 178,614 comments. Based on the results, DistilBERT achieved the best overall results with an F1-score of 0.73 and a good balance between precision of 0.68 and recall of 0.77 at a 0.70 threshold.

The results also highlight the impact of class imbalance on model performance. Which threshold tuning was found to be an effective technique to find balance between precision and recall in imbalanced datasets.

Future work may focus on improving the dataset further through more data augmentation techniques, as well as adding a validation dataset for better evaluation. Additional improvements can be applied like changing class weights and experimenting with hyperparameters.

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